

PGR 210 - Natural Language Processing

Lec9

By Huamin Ren

Huamin.ren@kristiania.no

Outline

- Learning the history of neural networks
- Stacking perceptrons
- Understanding backpropagation
- Implementing a basic neural network in Keras

Outline

- *Learning the history of neural networks*
- Stacking perceptrons
- Understanding backpropagation
- Implementing a basic neural network in Keras

Neural network

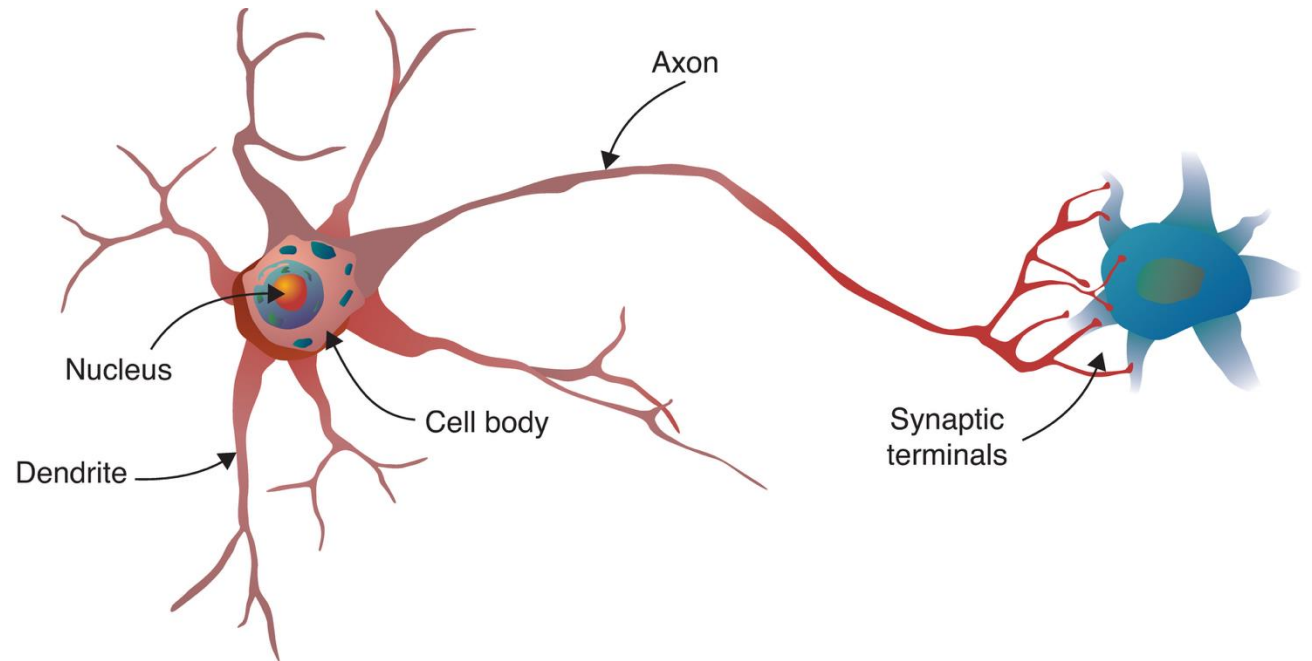
- Real neurons
- Artificial neurons
- Initializing the weights
- Deep networks
- Fully connected layers
- Tensors
- Preventing network collapse

Real neurons

In biology, the term neuron is applied to a wide variety of complex cells distributed throughout every human body.

These cells all have similar structure and behaviors, *but* they are specialized for many different tasks.

Neurons are sophisticated pieces of biology that use a mix of chemistry, physics, electricity, timing, proximity, and other means to perform their behaviors and communicate with one another.



Artificial neurons

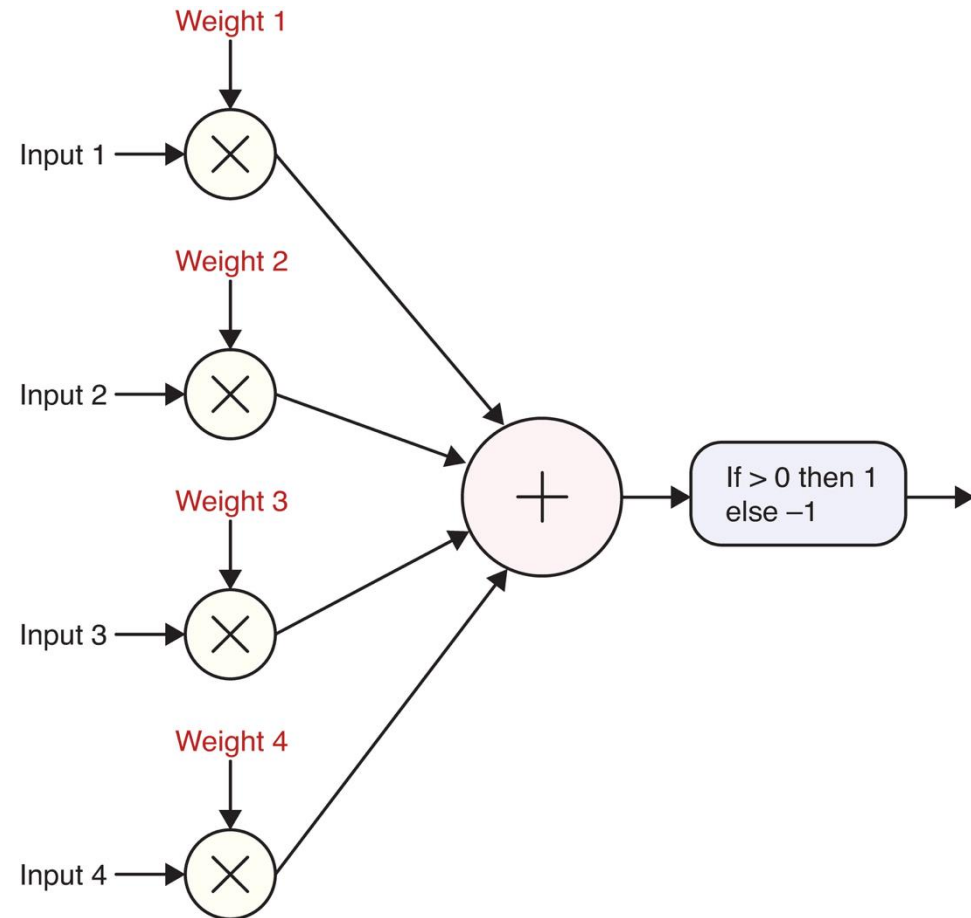
- The history of artificial neurons may be said to begin in 1943, with the publication of a paper that presented a massive simplified abstraction of a neuron's basic functions in mathematical form and described how multiple instances of this object could be connected into a network

McCulloch and Pitts 1943

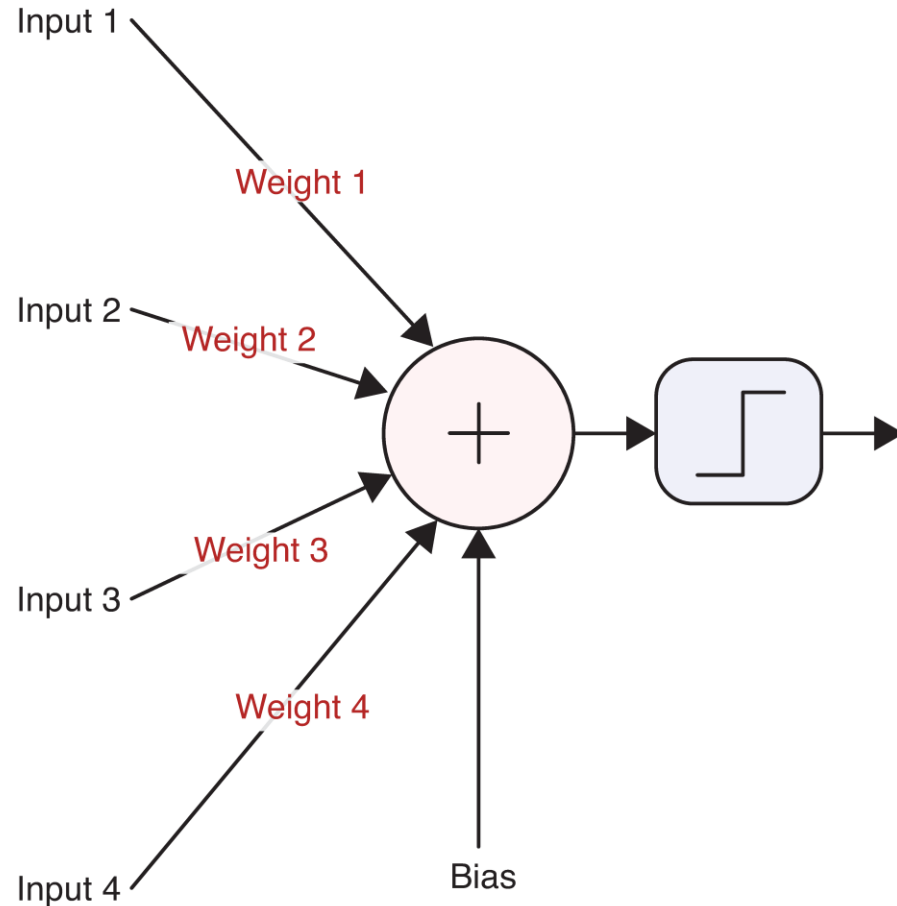
Outline

- Learning the history of neural networks
- **Stacking perceptrons**
- Understanding backpropagation
- Implementing a basic neural network in Keras

- Every input to a perceptron is represented by a single floating-point number (a weight)
- The results of these multiplications are all added together
- The results is compared to a threshold value; >0 , output >1 , otherwise, -1

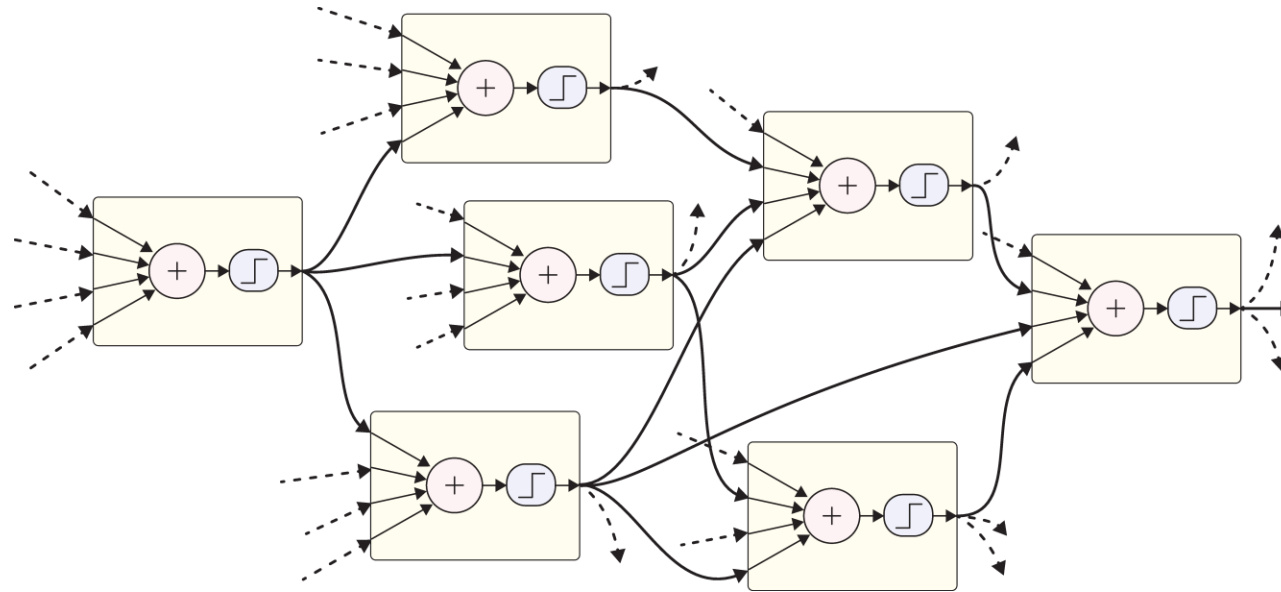


Modern Artificial Neurons

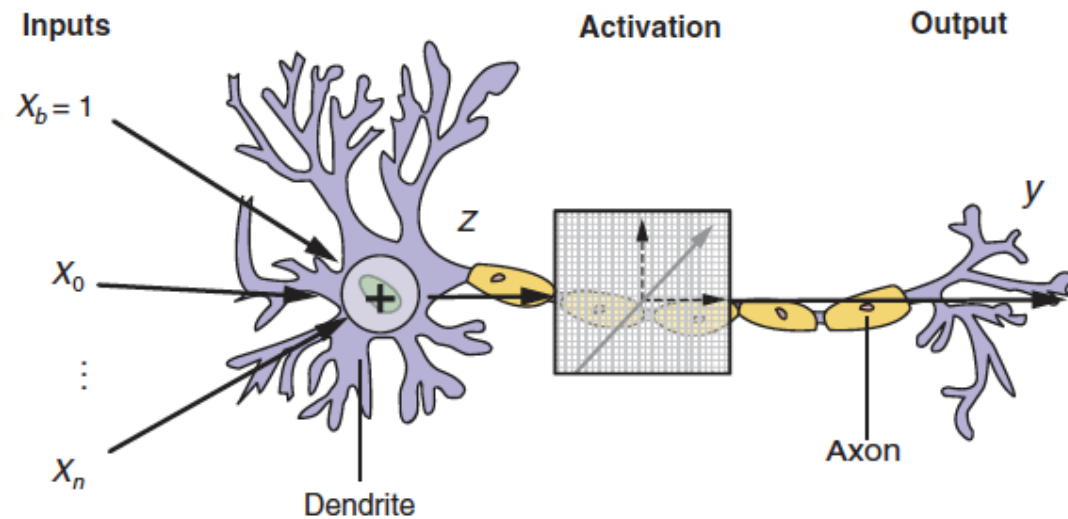


- A neuron is often drawn with the weights on the arrows

Modern Artificial Neurons



A piece of a larger network of artificial neurons. Each neuron receives its inputs from other neurons. The dashed lines show connections to and from outside this little cluster



A perceptron and a biological neuron

Read pages 156-159 in textbook (20 min): NLP...action

Initializing the weights

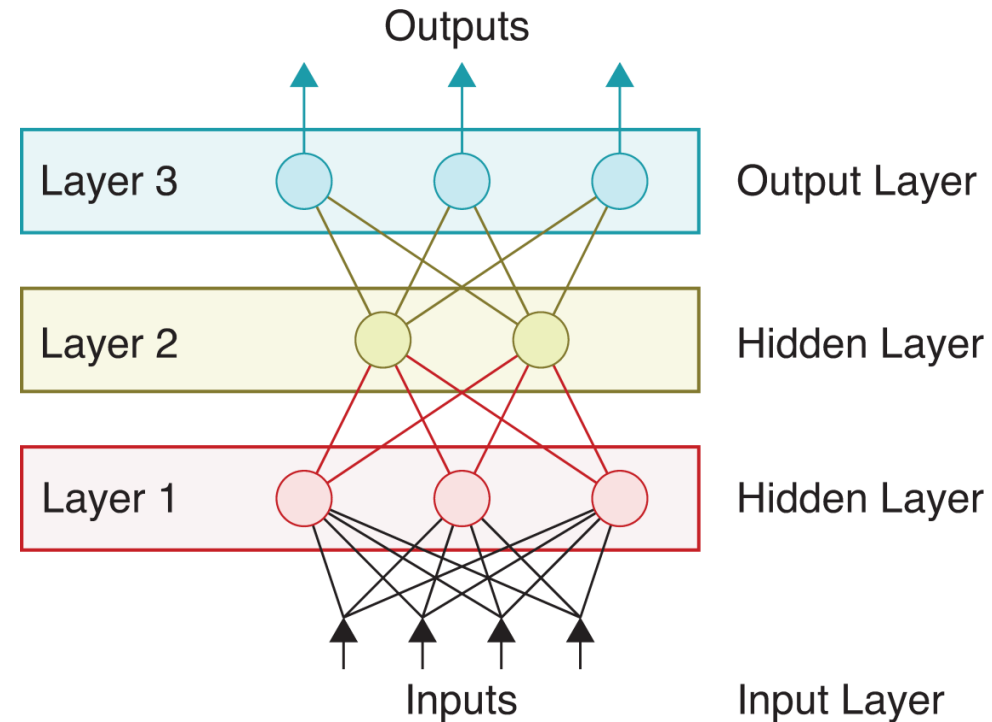
- Teaching a neural network involves gradually improving the weights
- The process begins when we assign initial values to the weights
- Researchers have developed theories for good ways to pick the initial values for the weights, e.g. The LeCun uniform

<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=726791>

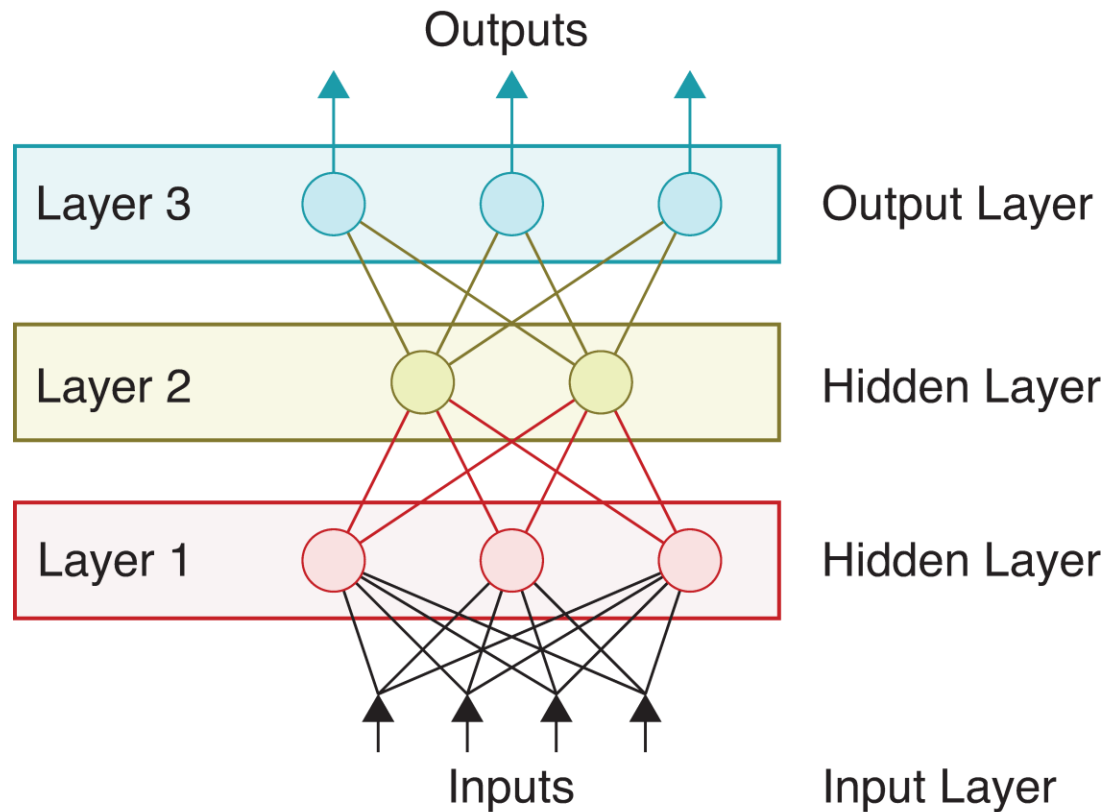
<https://www.nature.com/articles/nature14539>

Deep networks

- A network made of a series of layers that we often draw vertically



Deep networks

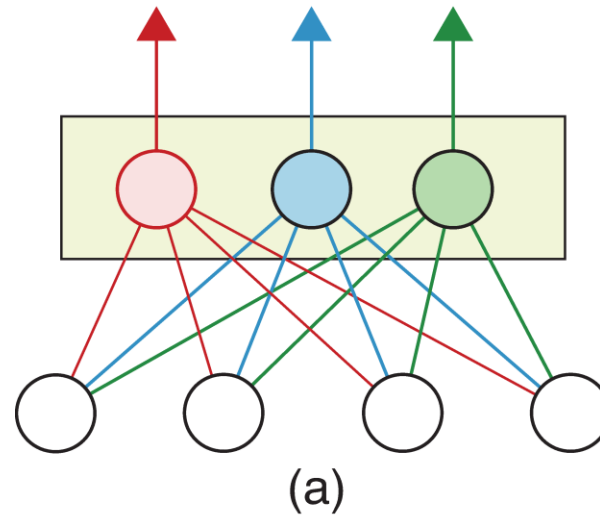


- The first layer: looks at the individual pixels;
- The second layer looks at groups of pixels;
- The next layer looks at groups of those than others;
- And so on.

- Early layers might notice that some pixels are darker than others
- Later layers might notice that a clump of pixels looks like an eye
- A much later layer might notice the collection of shapes that reveal that the whole image shows a tiger...

Fully connected layers

- FC, Or dense layers, is a set of neurons that each receive an input from every neuron on the previous layer



Outline

- Learning the history of neural networks
- Stacking perceptrons
- *Understanding backpropagation*
- Implementing a basic neural network in Keras

